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Entropy

Question

Give the definition of entropy for discrete random variables.

Explain the main properties of entropy.

Explain the entropy of binary random variables.

Definition of Entropy for Discrete Random Variables

Entropy is a fundamental measure in information theory that quantifies the **uncertainty** or **lack of information** associated with a discrete random variable.

Mathematical Definition

For a discrete random variable X taking values in an alphabet $\mathcal{X}$ with a probability mass function (pmf) $p_X(x) = \Pr(X = x)$, the entropy $H(X)$ is defined as:

$$H(X) = - \sum_{x \in \mathcal{X}} p_X(x) \log_2 p_X(x)$$

Unit: The logarithm is generally in base 2, giving the unit in **bits**. In base e , the unit is **nats**.

Intuitive Interpretation

- **High entropy** = high uncertainty, “spread out” distribution
- **Low entropy** = low uncertainty, “concentrated” distribution
- **Zero entropy** = no uncertainty (deterministic variable)

Entropy represents the **minimum average number of bits** required to encode the outcomes of X .

Main Properties of Entropy

1. Non-negativity

$$H(X) \geq 0$$

- Equality $H(X) = 0$ occurs if and only if X is **deterministic** (one outcome has probability 1, others 0)
- This means there is no uncertainty about X

2. Upper Bound: Maximum Entropy

For an alphabet of size $|\mathcal{X}| = M$, entropy is maximized when the distribution is **uniform**:

$$H(X) \leq \log_2 M$$

- Equality $H(X) = \log_2 M$ occurs if and only if $p_X(x) = \frac{1}{M}$ for all $x \in \mathcal{X}$
- In this case, uncertainty is maximal: every outcome is equally likely

3. Convexity/Concavity

Entropy is a **concave** function of the probability distribution $p_X(x)$.

This means that for two distributions p and q and for $\lambda \in [0, 1]$:

$$H(\lambda p + (1 - \lambda)q) \geq \lambda H(p) + (1 - \lambda)H(q)$$

Interpretation: Mixing two distributions increases uncertainty (or preserves it).

4. Additivity for Independent Variables

If X and Y are **independent**, then:

$$H(X, Y) = H(X) + H(Y)$$

More generally, for N i.i.d. (independent and identically distributed) variables:

$$H(X_1, X_2, \dots, X_N) = N \cdot H(X)$$

5. Conditioning Property

The entropy of a pair of variables satisfies:

$$H(X, Y) = H(X) + H(Y|X) = H(Y) + H(X|Y)$$

Where $H(Y|X)$ is the **conditional entropy**, measuring the uncertainty of Y given X .

6. Conditional Entropy is Less Than or Equal to Unconditional Entropy

$$H(Y|X) \leq H(Y)$$

- Equality holds if and only if X and Y are independent
- This means that knowing a variable (even partially) can only **reduce** the uncertainty about the other

7. Invariance Under Bijection

Entropy is **invariant** under relabeling of symbols. Only the **probabilities** matter, not the values themselves.

Entropy of Binary Random Variables

Definition of the Binary Entropy Function

For a binary random variable $X \in \{0, 1\}$ with:

- $\Pr(X = 1) = p$
- $\Pr(X = 0) = 1 - p$

The binary entropy, denoted $H_b(p)$ or simply $H(p)$, is defined as:

$$H_b(p) = -p \log_2 p - (1 - p) \log_2 (1 - p)$$

By convention, we set $0 \log_2 0 = 0$ (continuous limit).

Properties of the Binary Entropy Function

1. Symmetry

$$H_b(p) = H_b(1 - p)$$

The function is symmetric around $p = 0.5$.

2. Values at Boundaries

- $H_b(0) = 0$: if $p = 0$ or $p = 1$, the variable is deterministic
- $H_b(0.5) = 1$: maximum, uniform distribution over 2 symbols

3. Bell Shape

The function $H_b(p)$ has a characteristic “bell” shape:

- Increasing on $[0, 0.5]$
- Decreasing on $[0.5, 1]$
- Maximum of 1 bit at $p = 0.5$



Practical Interpretation

1. **Balanced Binary Source** ($p = 0.5$):
 - Maximum uncertainty: 1 bit per symbol
 - Example: fair coin

2. Unbalanced Binary Source ($p \neq 0.5$):

- Reduced uncertainty
- Example: asymmetric noise in transmission

3. Almost Deterministic Source ($p \approx 0$ or $p \approx 1$):

- Very low entropy
- High redundancy, good compressibility

Numerical Example

Probability p	Entropy $H_b(p)$	Interpretation
0.0 or 1.0	0.000 bits	No uncertainty
0.1 or 0.9	0.469 bits	Low uncertainty
0.2 or 0.8	0.722 bits	Moderate uncertainty
0.3 or 0.7	0.881 bits	High uncertainty
0.4 or 0.6	0.971 bits	Very high uncertainty
0.5	1.000 bits	Maximum uncertainty

Applications

1. **Data Compression:** Entropy gives the lossless compression limit
2. **Channel Coding:** Determines the capacity of binary channels
3. **Machine Learning:** Used in decision trees (information gain)
4. **Cryptography:** Measure of randomness in key generation

Summary

Concept	Description	Formula/Example
Definition	Measure of uncertainty of a random variable	$H(X) = -\sum p(x) \log_2 p(x)$
Non-negativity	Always non-negative	$H(X) \geq 0$
Maximum	Achieved for uniform distribution	$H(X) \leq \log_2 M$
Concavity	Mixing distributions increases entropy	$H(\lambda p + (1-\lambda)q) \geq \lambda H(p) + (1-\lambda)H(q)$

Concept	Description	Formula/Example
Binary Entropy	Special case for two outcomes	$H_b(p) = -p \log_2 p - (1-p) \log_2 (1-p)$
Binary Symmetry	Same uncertainty for p and $1-p$	$H_b(p) = H_b(1-p)$

Entropy is thus a central concept that quantifies information and uncertainty, with elegant mathematical properties and broad practical applications in data science, from communications to machine learning.